

Environmental Protection

The major environmental issues that have become a cause of concern are population growth, rapid and wasteful use of resources, loss of biodiversity and degradation of ecosystems, pollution of the water, air and soil, poverty, disparity in economic growth, global climate change, energy crisis and unsustainable growth trends. In order to protect the environment efforts have to be done at individual level as well as national and international level. The role of government and non-government organizations (NGOs) are equally important. Legislative measures are also very important in ensuring environmental protection. Environmental awareness, education and particularly, women education hold a significant place in promoting the spirit of environmental protection.

8.1 ROLE OF GOVERNMENT

Environmental protection can be done by formulating some guidelines, policies and laws by the government.

India is the first country in the world to have made provisions for the protection and conservation of environment in its Constitution. On 5th June, 1972, environment was first discussed as an item of international agenda in the **U.N. Conference on Human Environment** in Stockholm and thereafter **5th June** is celebrated all over the world as **World Environment Day**. Soon after the Stockholm Conference our country took substantive legislative steps for environmental protection. The Wildlife (Protection) Act was passed in 1972, followed by the Water (Prevention and Control of Pollution) Act 1974, the Forest (Conservation) Act, 1980, Air (Prevention and Control of Pollution) Act, 1981 and subsequently the Environment (Protection) Act, 1986.

Constitutional Provisions

Provisions for environmental protection in the Constitution of India were made within four years of Stockholm Conference, in 1976, through the 42nd amendment as follows:

Article 48-A of the Constitution provides: *“The state shall endeavour to protect and improve the environment and to safeguard forests and wildlife of the country.”*

Article 51A(g) provides: *“It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures.”*

Thus our Constitution includes environmental protection and conservation as one of our fundamental duties.

Some of the important Acts passed by the Government of India are discussed here. These Acts have the force of law and these environment laws are powerful tools for implementing environmental protection.

8.2 LEGAL ASPECTS

8.2.1 Wildlife (Protection) Act, 1972

The Act, a landmark in the history of wildlife legislation in our country, came into existence in 1972. Wildlife was transferred from State List to Concurrent List in 1976, thus giving power to the Central government to enact the legislation.

The **Indian Board of Wildlife (IBWL)** was created in 1952 in our country, which after the enactment of the Wildlife (Protection) Act actively took up the task of setting up wildlife national parks and sanctuaries. The major activities and provisions in the Act can be summed up as follows:

- (i) It defines the wildlife related terminology.
- (ii) It provides for the appointment of Wildlife Advisory Board, wildlife warden, their powers, duties etc.
- (iii) Under the Act, comprehensive listing of endangered wildlife species was done for the first time and prohibition of hunting of the endangered species was mentioned.
- (iv) Protection to some endangered plants like Beddome cycad, Blue vanda, Ladies slipper orchid, Pitcher plant etc. is also provided under the Act.
- (v) The Act provides for setting up of national parks, wildlife sanctuaries etc.
- (vi) The Act provides for the constitution of Central Zoo Authority.
- (vii) There is provision for trade and commerce in some wildlife species with license for sale, possession, transfer etc.
- (viii) The Act imposes a ban on the trade or commerce in scheduled animals (Fig. 8.1).
- (ix) It provides for legal powers to officers and punishment to offenders.
- (x) It provides for captive breeding programme for endangered species.

Several conservation projects for individual endangered species like lion (1972), tiger (1973), crocodile (1974) and brown antlered deer (1981) were started under this Act. The Act is adopted by all states in India except J & K, which has its own Act.

Some of the major drawbacks of the Act include mild penalty to offenders, illegal wildlife trade in J & K, personal ownership certificate for animal articles like tiger and leopard skins, no coverage of foreign endangered wildlife, pitiable condition of wildlife in mobile zoos and little emphasis on protection of plant genetic resources.



Fig. 8.1. A newspaper clipping advocating illegality of trading in scheduled animals and products.

8.2.2 Forest (Conservation) Act, 1980

This Act deals with the conservation of forests and related aspects. Except J & K, the Act is adopted all over India. The Act covers under it all types of forests including reserve forests, protected forests or any forested land irrespective of its ownership.

The salient features of the Act are as follows:

(i) The State government has been empowered under this Act to use the forests only for forestry purposes. If at all it wants to use it in any other way, it has to take prior approval of Central government, after which it can pass orders for declaring some part of the reserve forest for non-forest purposes (e.g. mining) or for clearing some naturally growing trees and replacing them by economically important trees (reforestation).

(ii) It makes provision for conservation of all types of forests and for this purpose there is an advisory committee which recommends funding for it to the Central government.

(iii) Any illegal non-forest activity within a forest area can be immediately stopped under this Act.

Non-forest activities include clearing of forest land for cultivation of any type of plants/crops or any other purpose (except re-afforestation). However, some construction work in the forest for wildlife or forest management is exempted from non-forest activity (e.g. fencing, making water-holes, trench, pipelines, check posts, wireless communication etc.).

1992 Amendment in the Forest Act

- In 1992, some amendments were made in the Act which made provisions for allowing some non-forest activities in forests, without cutting trees or limited cutting with prior approval of Central government. These activities are setting of transmission lines, seismic surveys, exploration, drilling and hydroelectric projects. The last activity involves large scale destruction of forests, for which prior approval of the Centre is necessary.
- Wildlife sanctuaries, national parks etc. are totally prohibited for any exploration or survey under this Act without prior approval of Central govt. even if no tree-felling is involved.

- Cultivation of tea, coffee, spices, rubber and plants which are cash-crops, are included under non-forestry activity and not allowed in reserve forests.
- Even cultivation of fruit-bearing trees, oil-yielding plants or plants of medicinal value in forest area need to be first approved by the Central govt. This is because newly introduced species in the forest area may cause an imbalance in the ecology of the forest. If the species to be planted is a native species, then no prior clearance is required.
- Tusser cultivation (a type of silk-yielding insect) in forest areas by tribals as a means of their livelihood is treated as a forestry activity as long as it does not involve some specific host tree like Asan or Arjun. This is done in order to discourage monoculture practices in the forests which are otherwise rich in biodiversity.
- Plantation of mulberry for rearing silkworm is considered a non-forest activity. The reason is same as described above.
- Mining is a non-forestry activity and prior approval of Central govt. is mandatory. The Supreme Court in a case *T.N. Godavarman Thirumulkpad Vs. Union of India (1997)* directed all on-going mining activity to be ceased immediately in any forest area of India if it had not got prior approval of Central government.
- Removal of stones, bajri, boulder etc. from river-beds located within the forest area fall under non-forest activity.
- Any proposal sent to Central govt. for non-forest activity must have a cost-benefit analysis and Environmental Impact Statement (EIS) of the proposed activity with reference to its ecological and socio-economic impacts.

Thus, the Forests (Conservation) Act has made ample provisions for conservation and protection of forests and prevent deforestation.

8.2.3 Water (Prevention and Control of Pollution) Act, 1974

It provides for maintaining and restoring the wholesomeness of water by preventing and controlling its pollution. ***Water pollution is defined as such contamination of water, or such alteration of the physical, chemical or biological properties of water, or such discharge as is likely to cause a nuisance or render the water harmful or injurious to public health and safety or harmful for any other use or to aquatic plants and other organisms or animal life.***

The definition of water pollution has thus encompassed the entire probable agents in water that may cause any harm or have a potential to harm any kind of life in any way.

The salient features and provisions of the Act are summed up as follows:

- (i) It provides for maintenance and restoration of quality of all types of surface and ground water.
- (ii) It provides for the establishment of Central and State Boards for pollution control.
- (iii) It confers them with powers and functions to control pollution.

The Central and State Pollution Control Boards are widely represented and are given comprehensive powers to advise, coordinate and provide technical assistance for prevention and control of pollution of water.

- (iv) The Act has provisions for funds, budgets, accounts and audit of the Central and State Pollution Control Boards.

(v) The Act makes provisions for various penalties for the defaulters and procedure for the same.

The main regulatory bodies are the Pollution Control Boards, which have been conferred the following duties and powers:

(A) Central Pollution Control Board (CPCB):

- It advises the Central govt. in matters related to prevention and control of water pollution.
- Coordinates the activities of State Pollution Control Boards and provides them technical assistance and guidance.
- Organizes training programs for prevention and control of pollution.
- Organizes comprehensive programs on pollution related issues through mass media.
- Collects, compiles and publishes technical and statistical data related to pollution.
- Prepares manuals for treatment and disposal of sewage and trade effluents.
- Lays down standards for water quality parameters.
- Plans nation-wide programs for prevention, control or abatement of pollution.
- Establishes and recognizes laboratories for analysis of water, sewage or trade effluent sample.

(B) The State Pollution Control Boards:

These boards also have similar functions to be executed at state level and are governed by the directions of CPCB.

- The Board advises the State govt. with respect to the location of any industry that might pollute a stream or a well.
- It lays down standards for effluents and is empowered to take samples from any stream, well or trade effluent or sewage passing through an industry.
- The State Board is empowered to take legal samples of trade effluent in accordance with the procedure laid down in the Act. The sample taken in the presence of the occupier or his agent is divided into two parts, sealed, signed by both parties and sent for analysis to some recognized lab. If the samples do not conform to the prescribed water quality standards (crossing maximum permissible limits), then 'consent' is refused to the unit.
- Every industry has to obtain consent from the Board (granted for a fixed duration) by applying on a prescribed proforma providing all technical details, along with a prescribed fee following which analysis of the effluent is carried out.
- The Board suggests efficient methods for utilization, treatment and disposal of trade effluents.

The Act has made detailed provisions regarding the power of the Boards to obtain information, take trade samples, restrict new outlets, restrict expansion, enter and inspect the units and sanction or refuse consent to the industry after effluent analysis.

While development is necessary, it is all the more important to prevent pollution, which can jeopardize the lives of the people. Installation and proper functioning of effluent treatment plants (ETP) in all polluting industries is a must for checking pollution of water and land. Despite certain weaknesses in the Act, the Water Act has ample provisions for preventing and controlling water pollution through legal measures.

8.2.4 The Air (Prevention and Control of Pollution) Act, 1981

Salient features of the act are as follows:

- (i) The Act provides for prevention, control and abatement of air pollution.
- (ii) In the Act, *air pollution has been defined as the presence of any solid, liquid or gaseous substance (including noise) in the atmosphere in such concentration as may be or tend to be harmful to human beings or any other living creatures or plants or property or environment.*
- (iii) Noise pollution has been inserted as pollution in the Act in 1987.
- (iv) Pollution Control Boards at the Central or State level have the regulatory authority to implement the Air Act. Just parallel to the functions related to Water (Prevention and Control of Pollution) Act, the Boards perform similar functions related to improvement of air quality. The Boards have to check whether or not the industry strictly follows the norms or standards laid down by the Boards under section 17, regarding the discharge of emission of any air pollutant. Based upon analysis report consent is granted or refused to the industry.
- (v) Just like the Water Act, the Air Act has provisions for defining the constitution, powers and function of Pollution Control Boards, funds, accounts, audit, penalties and procedures.
- (vi) Section 20 of the Act has provision for ensuring emission standards from automobiles. Based upon it, the State govt. is empowered to issue instructions to the authority incharge of registration of motor vehicles (under Motor Vehicles Act, 1939) that is bound to comply with such instructions.
- (vii) As per Section 19, in consultation with the State Pollution Control Board, the state government may declare an area within the state as “**air pollution control area**” and can prohibit the use of any fuel other than approved fuel in the area causing air pollution. No person shall, without prior consent of State Board operate or establish any industrial unit in the “air pollution control area”.

The Water and Air Acts have also made special provisions for appeals. Under Section 28 of Water Act and Section 31 of Air Act, a provision for appeals has been made. An **Appellate Authority** consisting of a single person or three persons appointed by the Head of the State, the Governor, is constituted to hear such appeals as filed by some aggrieved party (industry) due to some order made by the State Board within 30 days of passing the orders.

The Appellate Authority after giving the appellant and the State Board an opportunity of being heard, disposes off the appeal as expeditiously as possible.

8.2.5 The Environment (Protection) Act, 1986

The Act came into force on Nov. 19, 1986, the birth anniversary of our late Prime Minister Indira Gandhi, who was a pioneer of environmental protection issues in our country. The Act extends to whole of India. Some terms related to environment have been described as follows in the Act:

- (i) **Environment** includes water, air and land and the inter-relationships that exists among and between them and human beings, all other living organisms and property.
- (ii) **Environmental pollution** means the presence of any solid, liquid or gaseous substance present in such concentration, as may be, or tend to be, injurious to environment.
- (iii) **Hazardous substance** means any substance or preparation which by its physico-chemical properties or handling is liable to cause harm to human beings, other living organisms, property or environment.

The Act has given powers to the Central Government to take measures to protect and improve environment while the state governments coordinate the actions. The most important functions of Central govt. under this Act include setting up of:

- (a) The standards of quality of air, water or soil for various areas and purposes.
- (b) The maximum permissible limits of concentration of various environmental pollutants (including noise) for different areas.
- (c) The procedures and safeguards for the handling of hazardous substances.
- (d) The prohibition and restrictions on the handling of hazardous substances in different areas.
- (e) The prohibition and restriction on the location of industries and to carry on process and operations in different areas.
- (f) The procedures and safeguards for the prevention of accidents which may cause environmental pollution and providing for remedial measures for such accidents.

The power of entry and inspection, power to take sample etc. under this Act lies with the Central Government or any officer empowered by it.

For the purpose of protecting and improving the quality of the environment and preventing and abating pollution, standards have been specified under Schedule I-IV of Environment (Protection) Rules, 1986 for emission of gaseous pollutants and discharge of effluents/waste water from industries. These standards vary from industry to industry and also vary with the medium into which the effluent is discharged or the area of emission. For instance, the maximum permissible limits of B.O.D. (Biochemical Oxygen Demand) of the waste water is 30 ppm if it is discharged into inland waters, 350 ppm if discharged into a public sewer and 100 ppm, if discharged onto land or coastal region. Likewise, emission standards vary in residential, sensitive and industrial area. Naturally the standards for sensitive areas like hospitals are more stringent. It is the duty of the Pollution Control Board to check whether the industries are following the prescribed norms or not.

Under the **Environmental (Protection) Rules, 1986** the State Pollution Control Boards have to follow the guidelines provided under Schedule VI, some of which are as follows:

- (a) They have to advise the industries for treating the waste water and gases with the best available technology to achieve the prescribed standards.
- (b) The industries have to be encouraged for recycling and reusing the wastes.
- (c) They have to encourage the industries for recovery of biogas, energy and reusable materials.
- (d) While permitting the discharge of effluents and emissions into the environment, the State Boards have to take into account the assimilative capacity of the receiving water body.
- (e) The Central and State Boards have to emphasize on the implementation of clean technologies by the industries in order to increase fuel efficiency and reduce the generation of environmental pollutants.

Under the Environment (Protection) Rules, 1986 an amendment was made in 1994 for Environmental Impact Assessment (EIA) of various development projects. There are 29 types of projects listed under Schedule I of the rule which require clearance from the Central Government before establishing.

Others require clearance from the State Pollution Control Board, when the proposed project or expansion activity is going to cause pollution load exceeding the existing levels. The project proponent

has to provide EIA report, risk analysis report, NOC from State Pollution Control Board, commitment regarding availability of water and electricity, summary of project report/feasibility report, filled in a questionnaire for environmental appraisal of the project and comprehensive rehabilitation plan, if more than 1000 people are likely to be displaced due to the project.

Under the Environment (Protection) Act, 1986 the Central Government also made the Hazardous Wastes (Management and Handling) Rules, 1989. Under these rules, it is the responsibility of the occupier to take all practical steps to ensure that such wastes are properly handled and disposed off without any adverse effects. There are 18 hazardous waste categories recognized under this rule and there are guidelines for their proper handling, storage, treatment, transport and disposal which should be strictly followed by the owner.

The Environment (Protection) Act, 1986 has also made provision for environmental audit as a means of checking whether or not a company is complying with the environmental laws and regulations. Thus, ample provisions have been made in our country through law for improving the quality of our environment.

8.3 INITIATIVES BY NON-GOVERNMENTAL ORGANIZATIONS

(A) Role:

Non-government organizations (NGOs) can play a very important role in protection of environment because they act at the grass-root level. Simple laws cannot be enforced that effectively unless there is proper awareness amongst the masses. The NGOs can play a dual role:

(i) They can act as watch-dogs and advice the government about some local environmental issues of prime and urgent concern

(ii) They can interact with the people at grass-root level, sharing their problems and concern. These NGO's can act as a viable link between the two.

They can act both as an "Action group" and a "pressure group" by leading public environmental movements.

(B) Drawbacks:

(i) Quite often NGO's are found to work with vested interests, without a genuine concern for the environment

(ii) They may get politically motivated. In such cases the public feels cheated and the enthusiasm to protect the environment is totally lost.

Nonetheless, all over the world the number of NGOs is increasing day by day and several initiatives have been taken by them for protection of environment in different ways.

(C) Initiatives:

Some major initiatives taken by NGO's are discussed here:

(i) **Dasholi Gram Swarajya Mandal** in Gopeshwar is known for the well known "Chipko Movement" for protecting the trees. Sh. Sunderlal Bahuguna's name is now synonymous with this movement, who led this movement in Uttarakhand against tree-felling.

(ii) **Kalpavriksh** is known for the “Narmada Bachao Andolan” headed by Medha Patekar, raising the major environmental issues associated with the Sardar Sarovar dam on the river, particularly the issue of displacement and rehabilitation of the natives/outsees.

(iii) **Centre for Science and Environment (CSE)** have played a significant role in preparing ‘Citizen’s Report’ and have taken up various prime issues in a scientific way. Pesticide levels in cola drinks exceeding the maximum permissible limits has sensitized the people all over the country.

(iv) **Bombay Natural History Society (BNHS)** is one of the oldest NGOs of India who have worked for the protection of wildlife.

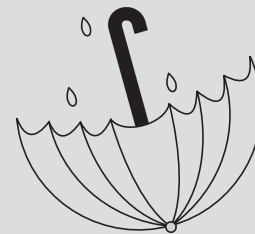
(v) **Tarun Bharat Sangh**, Rajasthan, under the leadership of Sh. Rajendra Singh has done a remarkable job of harvesting rainwater by constructing check-dams, who was honoured with prestigious Magsaysay award for his work.

Water Conservation Efforts by CSE (case study)

Rainwater Harvesting

Late Sh. Anil Agarwal, Founder Director, and CSE tried to draw attention of general public towards conservation of water:

“The Real Green Revolution is about rainwater harvesting. Let us catch water where it falls. Let it transform human lives. Let it change social existence. If this happens, the world will be transformed. The world will merely be an agglomeration of ecological-rainwater harvesting-democracies.”



In the recent years, there has been a revival of traditional water harvesting systems in various parts of India. Facing huge water crisis, rural communities have revived and created new water harvesting systems such as checkdams, *johads*, rooftop runoffs and other structures for harvesting every drop of water.

The Centre for Science and Environment (CSE) has launched a campaign to spread awareness about community-based rainwater harvesting techniques, and has identified 16 model projects in Delhi which are being monitored regularly. The systems are successfully running under different geographical and geological condition. In the Mira Model School, for example, it has been constructed on a sedimentary terrain while the Shri Ram School project is on a hilly terrain with different forms of rainwater harvesting *i.e.*, rooftop harvesting and surface water harvesting.

Paani Yatra

It is another novel way by which the CSE has prompted people to rediscover water by undertaking a trip to different villages where local communities have harvested rainwater and used it wisely. In the process, they have succeeded in achieving a dramatic revival of local ecology and economy. Paani yatra is a means to arouse awareness amongst individuals and organisations for water self-reliance, popularly called as **Jal Swaraj**, which offers a unique opportunity to masses to attain water literacy.

Sunita Narain, Director, CSE, was awarded the “World Water Prize” in Stockholm in Aug. 2005 in recognition of the work done by CSE for water conservation.

Greenpeace is a world-wide NGO working for the protection of environment. This organization believes in peaceful mass movements for environmental protection. There is **Sea-shepherd**, another NGO, which is determined to stop killing of marine animals. This NGO even resorted to aggressive and violent means to achieve their goals.

8.4 ENVIRONMENTAL EDUCATION

8.4.1 Need for Environmental Education

Environmental education or environmental literacy is something that every person should be well versed with. The principles of ecology and fundamentals of environment can really help create a sense of earth-citizenship and a sense of duty to care for the earth and its resources and to manage them in a sustainable way so that our children and grand children too inherit a safe and clean planet to live on.

Following the Supreme Court directives (in M.C. Mehta Vs. Union of India, 1988) environmental education has been included in the curriculum right from the school stage to college/university level. The prime objective of the same is to make everyone environment literate. The environment belongs to each one of us and our actions affect the environment. When the environment gets degraded it affects our health, well-being and our future. So we have a right to know the *a b c* of environment and also have a right to safe and clean environment.

8.4.2 Need for Value-based Environmental Education

Let us now see how environmental education be made value-oriented.

1. **Human values.** Preparation of textbooks and resource materials about environmental education can play an important role in building positive attitudes about environment. The basic human value '**man in nature**' rather than '**nature for man**' needs to be infused through the same.

2. **Social values.** Love, compassion, tolerance and justice which are the basic teachings of most of our religions need to be woven into environmental education. These are the values to be nurtured so that all forms of life and the biodiversity on this earth are protected.

3. **Cultural and religious values.** These are the values enshrined in Vedas like "*Dehi me dadami te*" i.e. "you give me and I give you" (Yajurveda) emphasize that man should not exploit nature without nurturing her. Our cultural customs and rituals in many ways teach us to perform such functions as would protect and nurture nature and respect every aspect of nature, treating them as sacred, be it rivers, earth, mountains or forests.

4. **Ethical values.** Environmental education should encompass the ethical values of earth-centric rather than human-centric world-view. The educational system should promote the earth-citizenship thinking. Instead of considering human being as supreme we have to think of the welfare of the earth.

5. **Global values.** The concept that the human civilization is a part of the planet as a whole and similarly nature and various natural phenomena over the earth are interconnected and inter-linked with special bonds of harmony. If we disturb this harmony anywhere there will be an ecological imbalance leading to catastrophic results.

6. Spiritual values. Principles of self-restraint, self-discipline, contentment, reduction of wants, freedom from greed and austerity are some of the finest elements intricately woven into the traditional and religious fabric of our country. All these values promote conservationism and transform our consumeristic approach.

The above-mentioned human values, socio-cultural, ethical, spiritual and global values incorporated into environmental education can go a long way in attaining the goals of sustainable development and environmental conservation. Value-based environmental education can bring in a total transformation of our mindset, our attitudes and our lifestyles.

8.4.3 Approaches to Impart Environmental Education

Environmental education needs to be imparted through formal and informal ways to all sections of the society. Everyone needs to understand it because 'environment belongs to all' and 'every individual matters' when it comes to conservation and protection of environment.'

Various stages and methods that can be useful for raising environmental awareness in different sections of the society are as follows:

(i) **Among students through formal education:** Environmental education must be imparted to the students right from the childhood stage. It is a welcome step that now all over the country we are introducing environmental studies as a subject at all stages including school and college level, following the directives of the Supreme Court.

(ii) **Among the masses through mass-media:** Media can play an important role to educate the masses on environmental issues through articles, environmental rallies, plantation campaigns, street plays, real eco-disaster stories and success stories of conservation efforts. TV serials like *Virasat*, *Race to Save the Planet*, *Heads and Tails*, *Terra-view*, *Captain planet* and the like have been effective in propagating the seeds of environmental awareness amongst the viewers of all age groups. (Plate VI, VII).

(iii) **Among the planners, decision-makers and leaders:** Since this elite section of the society plays the most important role in shaping the future of the society, it is very important to give them the necessary orientation and training through specially organized workshops and training programmes.

Publication of environment-related resource material in the form of pamphlets or booklets published by Ministry of Environment & Forests can also help in keeping this section abreast of the latest developments in the field.

8.5 WOMEN EDUCATION

8.5.1 Status of Women

In many developing countries, women do not enjoy the same rights and opportunities as men do. Women make roughly half of the world's population, but as per WHO estimates they carry out 67% of total work, but receive only 10% of world's income. These figures clearly show the wide gender bias. Representation of women in government, world's parliaments, higher education, high managerial

positions is gradually increasing, but still it is far behind that of men. It is an irony that women make up about 70% of the world's poor and about 67% of the illiterate people. Women, especially in developing countries, spend most of their time growing food, gathering fuelwood, bringing water and doing domestic work and child care.

8.5.2 Need for Women Education

- (i) Educating the women will empower them to seek gender equality in the society.
- (ii) Women will be able to earn that would raise their economic condition and their status in the society.
- (iii) They will be aware about the advantages of small and planned family and this will be a big step towards achieving stabilized population goals.
- (iv) It has been reported that the single most important factor affecting high total fertility rates (TFR) is the low status of women in many societies. Women education will help increase the age of marriage of women and they would tend to have fewer, healthier children who would live longer.
- (v) Women on being educated would be able to rear their children in a better way, leading to their good health and provide them with better facilities.
- (vi) Women are also the victim of capitalism and development. Due to some development activity like dam building or mining, they get debilitated. The menfolk get some compensation and migrate to towns in search of some job while women are left behind to look after the family with little resources. They are compelled to take up some marginalized work, which is highly unorganized and often socially humiliating. Women education can greatly help restore their settlement and dignity.
- (vii) Education of women would mean narrowing down of social disparities and inequities. This would automatically lead to sustainable development.

I. QUESTIONS

1. Discuss the salient features of (a) Wildlife (Protection) Act, 1972 (b) Forest (Conservation) Act, 1980.
2. How do you define pollution as per Water (Prevention and Control of Pollution) Act, 1974 ? What are the salient features of the Act ?
3. Who has the authority to declare an area as "air pollution control area" in a state under the Air (Prevention and Control of Pollution) Act, 1981 ? When was noise inserted in this act ?
4. Why do we refer to Environmental Protection Act, 1986 as an Umbrella Act ? Discuss the Major Environmental Protection Rules, 1986.
5. What are the major limitations to successful implementation of our environmental legislation ?
6. What are the different methods to propagate environmental awareness in the society ?
7. Discuss the role of NGO's in environmental protection giving examples.
8. How can environmental education be made value oriented?
9. What are the methods of imparting environmental education?
10. Discuss the importance of women education.

II. OBJECTIVE TYPE QUESTIONS

(A) FILL IN THE BLANKS

1. The first country in the world to make provisions for environmental protection in its constitution is
2. Act provides for setting up of national parks and wildlife sanctuaries.
3. Noise has been included as pollution in the air (Prevention and control of pollution) Act, 1981 in the year
4. Any appeals filed by the aggrieved industrial unit against the State Pollution Control Board are heard and decided by authority, under the Water and Air Acts.
5. Environmental Protection Act came into force on 1986 the birth anniversary of Smt. Indira Gandhi.
6. There are types of projects which require clearance from central government before establishment.
7. Narmada Bachao Andolan has been launched by the NGO named
8. Environmental education has been made compulsory by the Supreme Court based upon the PIL filed by

(B) CHOOSE THE CORRECT ANSWER

1. Rajendra Singh of “Tarun Bharat Sangh” was awarded Magsaysay Award for his work on
 (a) Water conservation (b) Social forestry
 (c) Clean technology (d) Popularisation of solar energy.
2. Which article in constitution recognizes environmental protection as one of the fundamental duties of every citizen of India ?
 (a) Article 42 (b) Article 48A
 (c) Article 51A(g) (d) Article 52.
3. As per the Forest Act, cultivation of which of the following is a non-forest activity
 (a) Tea (b) Rubber
 (c) Mulberry (d) All of these.
4. Which of the following NGO's is associated with 'Chipko Movement' ?
 (a) Kalpavriksh (b) Srishti
 (c) Dasholi Gram Swarajya Mandal
 (d) Green Peace.
5. The sensitizing issue of pesticide contamination of cola drinks was brought into limelight by the NGO
 (a) Centre for Science and Environment
 (b) Bombay Natural History Society
 (c) Kerala Sastra Sahitya Parishad
 (d) Kalpavriksh.

(C) WRITE TRUE OR FALSE

1. The Wildlife (Protection) Act, 1972 is adopted all over India including Jammu & Kashmir.
(True/False)
2. For any non-forest activity in the forest prior approval of State Govt. is necessary, as per the 1992 amendment in Forest Act.
(True/False)
3. The definition of water pollution as per Water (Prevention and Control of Pollution) Act, 1974 includes not only the agents causing harm to any form of life but also those which have a likelihood of doing so.
(True/False)
4. Central and State Pollution Control Boards were established in 1996 under the Environmental (Protection) Act.
(True/False)
5. One of the major drawbacks of the Forest Conservation Act, 1980 is very poor community participation in it.
(True/False)
6. Paani Yatra is a way by which the people visit different villages to know and revive the local rain water harvesting technologies.
(True/False)
7. For environmental protection we should emphasize on human-centric world view in environmental education.
(True/False)
8. As per W.H.O. estimates the women carryout 2/3 of the total work and receive 50% of the world's income.
(True/False)